

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc Physics Examination April-May 2018

PS953 Solid State Physics -II

Date: 01-05-2018, Day: Tuesday Time: 10.00 AM To 10.20 AM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q-1 Choose the correct answer for the following questions

[10]

1. If the magnetic moment of substance is zero, the substance is -----.
 - (a) paramagnetic
 - (b) diamagnetic
 - (c) ferromagnetic
 - (d) ferrimagnetic
2. What type materials are required to make transformer cores?
 - (a) Soft magnetic materials
 - (b) Hard magnetic materials
 - (c) Superconducting materials
 - (d) Diamagnetic materials
3. The Kronig-Penney model assumes that:
 - (a) The potential energy of an electron to have the form of periodic array of square wells
 - (b) The potential energy of a proton to have the form of periodic array of square wells
 - (c) The potential energy of a neutron to have the form of periodic array of square wells
 - (d) All of these
4. Order parameter _____ with increase in temperature
 - (a) increases
 - (b) decreases
 - (c) does not change
 - (d) abruptly increases

5. Mesomorphic state refers to:
 - (a) Solid state
 - (b) Liquid state
 - (c) Liquid crystal state
 - (d) Gaseous state
6. The width of energy gap of a superconductor is maximum at:
 - (a) 0K
 - (b) T_c
 - (c) Room temperature
 - (d) Critical magnetic field
7. In Kronig –Penney model, as the scattering power of the potential barrier, $P \rightarrow \alpha$, then the allowed energy bands:
 - (a) reduces to single energy levels
 - (b) reduces to smaller bands
 - (c) increases to bigger bands
 - (d) no change
8. In superconducting state:
 - (a) Entropy increases and thermal conductivity decreases
 - (b) Entropy and thermal conductivity decreases
 - (c) Entropy and thermal conductivity increases
 - (d) Entropy decreases and thermal conductivity increases
9. Liquid crystal molecules possess:
 - (a) Translational order
 - (b) Vibrational order
 - (c) Intrinsic order
 - (d) Orientational order
10. In Kronig –Penney model, if there is no potential barrier, then
 - (a) There are forbidden energy regions
 - (b) There are no forbidden energy regions
 - (c) All values of energy are not allowed
 - (d) There is periodic dependence of E on k

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IV Semester of MSc Physics Examination April-May 2018

PS953 Solid State Physics -II

Date: 01-05-2018, Day: Tuesday Time: 10.20 AM To 12.00 Noon Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed **10**

1. What is a SQUID and how can it be used to measure a magnetic field? 2
2. Explain the hysteresis curve of ferromagnetic materials. Explain Hard and soft magnetic materials. 2
3. What is a quasicrystal? Explain with suitable examples. 2
4. Explain four major areas of technology where nanotechnology can play important roles. 2
5. A superconducting material has a critical temperature of 3.7K, and a magnetic field of 0.0306 tesla at 0K. find the critical field at 2K. 2

SECTION - II

Q-III Answer the following questions as directed (any five) **15**

1. What are cooper pairs and how are they formed? How are cooper pairs kept together? 3
2. Explain the important property changes that occur in materials when they change from normal to superconducting state. 3
3. What is a peculiar feature of nano-structures distinguished them from micro- and macro-structures? Explain in terms of size effects. 3
4. Discuss the Kronig-Penney Model to describe the behavior of an electron in a periodic potential. 3
5. What are Liquid Crystals? What are the parameters which describe the liquid crystalline structure? 3
6. How do diamagnets differ fundamentally from paramagnets and ferromagnets? Explain Langevin theory of paramagnetism. 3